WORKSHOP REPORT
SEED ACCELERATOR MEET 2.0

Organized by

International Rice Research Institute (IRRI)

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Lastly, we extend heartfelt thanks to all participants—including representatives from private seed companies and farmers' producer organizations—for their active engagement and valuable contributions. Your collective insights have meaningfully shaped the dialogue on rice breeding strategies and seed system priorities.

Key Participating Institutions

अटारी ATARI GUWAHATI ZONE-VI

अटारी ATARI KOLKATA ZONE-V

यू पी सीड

सत्यमेव जयते Government of Assam

ODISHA STATE SEEDS CORPORATION LTD

ASSAM SEEDS CORPORATION LIMITED

महाबीज

ADVANTA Modern Science - Traditional Values

Harlal SEEDS Prosperity (हर) Revolution (लाल)

Mahindra AGRI SOLUTIONS

NUZIVEEDU SEEDS

RALLIS INDIA LIMITED A TATA Enterprise

TRIMURTI Plant Sciences

Star agriseeds बुवाई से कटाई तक

Eagle Let's grow!

VNR SEED GOOD SEED BETTER YIELD

Siri SIRI SEEDS Quality for Prosperity

Srikar Seeds

SIGNET SEEDS

CSA CARING FOR THOSE WHO FEED THE NATION SINCE 2004

JAYA SEEDS

POOHAR AGRO PRODUCER COMPANY LIMITED

Abbreviations and Acronyms

AICRIP	All India Coordinated Rice Improvement Programme
ASOCA	Assam Seed and Organic Certification Agency
ATARI	Agricultural Technology Application Research Institutes
BS	Breeder seed
CSISA	Cereal Systems Initiative for South Asia
CVRC	Central Variety Release Committee
DELS-R	Direct-seeded Early-duration Long Slender soft grain rice for Rainfed ecosystems
DMeLS-R	Direct-seeded Medium-duration Long Slender soft grain rice for Rainfed ecosystems
EGS	Early-Generation Seed
FPC	Farmer Producer Company
FPO	Farmer Producer Organization
HYVs	High-Yielding Varieties
ICAR	Indian Council of Agricultural Research
ISARC	IRRI South Asia Regional Centre
KVK	Krishi Vigyan Kendra
MSP	Minimum Support Price
NARES	National Agricultural Research and Education System
NGO	Non-Government Organization
NSC	National Seeds Corporation
OFT	On-Farm Trial
OSSC	Odisha State Seed Corporation
SOP	Standard Operating Procedure
SRR	Seed Replacement Rate
SSC	State Seeds Corporation
SVRC	State Variety Release Committee
TEMS-I/R	Transplanted Early-duration Medium Slender soft grain rice for Irrigated/Rainfed ecosystems
TLaSF-R	Transplanted Late-duration Small Firm grain type rice for Rainfed ecosystems
TLS	Truthfully Labelled Seed
TMeLS-R	Transplanted Medium-duration Long Slender soft grain type rice for Rainfed ecosystems
UBVN	Uttar Pradesh Beej Vikas Nigam
VRR	Varietal Replacement Rate

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Executive Summary

The Seed Accelerator Meet 2025, convened by the IRRI South Asia Hub, served as a dynamic platform to deepen strategic collaborations and drive innovation across rice breeding and seed systems in India. Against the backdrop of growing climate variability and shifting market preferences, the workshop brought together a diverse coalition of stakeholders committed to transforming the rice seed landscape through accelerated varietal turnover and inclusive seed delivery models.

The workshop centered on IRRI's One Rice Breeding and Seed System Strategy, co-developed with NARES, which prioritizes demand-driven, market-segment-based product development, on-farm validation, and coordinated scaling efforts. Discussions focused on bridging persistent gaps in varietal adoption and improving the accessibility of high-performing, climate-resilient, and market-preferred rice varieties for farmers across India.
Participants from ICAR-ATARIs, state seed corporations, private seed enterprises, and farmer producer organizations converged to:

- Strategize on pathways to replace outdated rice varieties with superior alternatives;
- Strengthen forward linkages between breeders, seed agencies, and market actors;
- Share evidence and tools for varietal positioning and impact tracking;
- Explore collaborative models for ensuring that high-quality seeds reach diverse farming communities; and
- Address systemic bottlenecks hindering seed system responsiveness and efficiency.

Special emphasis was placed on harnessing multi-stakeholder networks and data-driven decision-making to guide product selection, positioning, and adoption. The workshop also facilitated cross-learning among seed system accelerators, promoting synergy between public and private sector actors to ensure that farmers can access seeds that are both agronomically suitable and commercially viable.

By reinforcing partnerships and aligning efforts across the value chain, the Seed Accelerator Meet 2025 set a renewed agenda for inclusive and sustainable seed system transformation. It reaffirmed the essential role of collaborative platforms in fostering innovation, equity, and resilience in rice production systems, ultimately contributing to enhanced food security, farm profitability, and rural livelihoods.

Venue and Participants

The Seed Accelerator Meet 2025 was convened at the International Rice Research Institute (IRRI) South Asia Regional Hub, located within the campus of the ICRISAT headquarters in Hyderabad, India. The city of Hyderabad was a strategically chosen venue as it is a major hub for the seed industry in South Asia. The city is home to numerous global and domestic seed companies, cutting-edge agricultural R&D institutions, and an ecosystem that fosters innovation and partnerships in seed systems and agribusiness. Hosting the event in Hyderabad provided participants with direct access to key players shaping the future of seed systems in the region.

The Indian Council of Agricultural Research - Agricultural Technology Application Research Institute (ICAR-ATARI) plays a pivotal role in coordinating and scaling agricultural innovations through the Krishi Vigyan Kendras (KVKs) network, and was invited to the Seed Accelerator Meet for its critical contributions in linking grassroots technologies with institutional and market ecosystems. The opening session featured a distinguished panel of experts and thought leaders who set the tone for the workshop with their insights into research, innovation, and partnerships in seed systems:

- Dr. Pradip Dey, Director, ICAR ATARI, Kolkata (Zone-V)
- Dr. Satish Pareek, R&D Lead, Advanta Seeds, Hyderabad
- Dr. Swati Nayak, Seed Systems Lead - South Asia, IRRI
- Dr. Vikas Kumar Singh, Regional Breeding Lead - South Asia, IRRI
- Dr. Mosharaf Hossain, Lead Specialist - Seed System and Product Management, IRRI
- Dr. Linga Reddy Gutha, Senior Manager - Business Development, IRRI

The opening session was also enriched by a virtual address from Dr. Sankalp Bhosale, Interim Director of IRRI's Rice Breeding Innovations, who shared insightful remarks on the evolving rice breeding priorities across South Asia.

The workshop brought together a dynamic and diverse group of participants, representing various nodes of the seed value chain. Stakeholders included:

- ICAR ATARIs (Kolkata, Guwahati and Kanpur) and state seeds corporations (SSCs) of Assam, Maharashtra, Odisha, and Uttar Pradesh, reflecting public sector engagement in seed value chain development
- Private seed companies, headquartered in key agri-business corridors such as Hyderabad, Delhi NCR, and Punjab

- Farmer producer organizations (FPOs) and farmer producer companies (FPCs) from Assam, Odisha, and Uttar Pradesh, representing the grassroots perspective on seed access, adoption, and demand-driven innovation

This rich blend of public, private, and community-based actors enabled productive discussions on seed system acceleration, public-private partnerships, varietal deployment, and farmer-centric seed strategies.

Operational and logistical support for the successful conduct of the workshop was provided by Dr. Neeraj Tyagi, Dr. Anirban Nath, Ms. Ankita Tiwari, Ms. Subhasmita Mohapatra, and Dr. Challa Venkateshwarlu, whose contributions ensured a seamless and productive event experience.

For a detailed list of participants and their affiliations, please refer to Appendix 1.

Strategic Insights from the Opening Session

The opening session of the Seed Accelerator Meet 2025 began with insightful remarks from the panel of experts. They asserted the strategic role of farmer-centric seed systems, research-driven varietal innovation, and multi-stakeholder partnerships in accelerating access to quality seeds.

Dr. Swati Nayak (Scientist and South Asia Seed System Lead, IRRI)
Dr. Swati Nayak expressed her gratitude to all KVKs and NGO (non-government organization) partners for their contributions in generating substantial evidence on improved rice varieties and their performance under farmer-managed practices. She noted that the objective of the Seed Accelerator Meet was to present these findings in an open forum, thereby increasing awareness among institutional stakeholders about the availability and performance of new varieties.

Dr. Nayak reiterated the crucial role played by ICAR-ATARIs and their extensive network of KVKs in the seed dissemination ecosystem, and stressed the importance of private seed companies in this framework. She stated that the event would provide updates on new products, varietal replacements, and varietal turnover trends. She also indicated that findings from on-farm trials (OFTs), conducted across various market segments and states, would be shared. Furthermore, she mentioned the annual update of IRRI's product diary, which highlights promising rice varieties that have undergone multiple field-level evaluations backed by robust evidence.

She informed the participants that IRRI would facilitate early-generation breeder seed (BS) linkages, and a requirement form would be shared with stakeholders to capture demand for different varieties. IRRI would then assist in connecting interested parties with relevant breeding resources, aiming to support the scaling of new products through collaboration with seed multipliers.

Dr. Nayak also noted that the session would conclude with a focused group discussion on the challenges faced by different institutional stakeholders, including ATARIs, private seed companies, and FPOs/FPCs, in the seed value chain, and would explore strategies for improvement.

She explained that the Seed Accelerator Meet was envisioned as a platform to foster public-private partnerships, enabling the dissemination of publicly bred high-yielding varieties (HYVs) to farmers through diverse distribution networks. The session would also include a presentation by Dr. Linga Reddy Gutha, Senior Manager-Business Development, IRRI Tech Transfer, on collaboration opportunities with IRRI in research and development, varietal line sharing, operational modalities, and subscription options.

In closing, Dr. Nayak mentioned the importance of making this an annual event to collectively review OFT outcomes, identify promising varieties, and strategize on scaling efforts. She stressed that the focus should not be on the number of products but on breakthrough varieties that can significantly benefit farmers. She encouraged participants to share their experiences and feedback to enhance IRRI's support going forward.

Dr. Sankalp Bhosale (Interim Director – Rice Breeding Innovations, IRRI)

Dr. Sankalp Bhosale extended a warm welcome to all participating partners from public institutions, private companies, and civil society organizations who are engaged in various aspects of the rice seed value chain, including commercialization, scaling, and systemic transformation.

He emphasized that one of the major challenges facing the breeding community is identifying the key drivers behind the adoption of new rice varieties. He posed a critical question regarding the average age of rice varieties in cultivation and observed that a comparison with crops like wheat, where CGIAR and national systems have stronger integration, shows rice falling behind. He noted that preliminary analysis suggests the average age of wheat varieties is nearly one-third that of rice, potentially due to differences in breeding strategies, institutional collaboration, and market delivery systems.

Dr. Bhosale remarked that despite these challenges, rice continues to be the most significant agricultural commodity in South Asia, both in terms of production and exports. However, he pointed out that the opportunity costs of delayed adoption of improved rice varieties are high, and that collaborative efforts between national systems and CGIAR institutions are essential to overcome these barriers.

He noted that selecting appropriate germplasm and ensuring its inclusion in the seed value chain is not straightforward, largely due to gaps in data on varietal performance, market acceptance, stress tolerance, and other agronomic traits. He urged national and state-level partners, including representatives from ATARI and SSCs, to collaborate with IRRI in generating robust evidence to better understand and catalyze the drivers of varietal adoption.

Dr. Bhosale also acknowledged the presence of private sector partners and acknowledged the importance of their role, particularly in hybrid rice development. He expressed IRRI's willingness to further engage with small and medium enterprises (SMEs) working with inbred rice varieties.

In conclusion, he conveyed his sincere appreciation for the continued support extended to IRRI's seed systems and breeding teams. He expressed hope that, through ongoing partnerships, stakeholders would be able to generate evidence that empowers farmers, informs policy, and fosters more informed decision-making across the rice sector.

Dr. Vikas Kumar Singh (Regional Breeding Lead – South Asia, IRRI)

Dr. Vikas Kumar Singh acknowledged that while a considerable amount of genetic material and data is being generated by the breeding and seed systems teams, access to this information remains limited. He pointed out that, in contrast, wheat germplasm tends to be more widely available. He emphasized that despite the scale of the breeding network, much of the valuable data is still not easily accessible to all stakeholders.

He shared that the breeding and seed systems teams have been working to assess the impact of newly developed products, evaluating how these improved varieties perform compared to checks and existing farmer-preferred or mega varieties. The team is also tracking the extent of the area covered under these new genetics. Additionally, over the past decade, the breeding team has conducted a comprehensive analysis to examine which materials have been developed, how they align with different market segments, and their placement among breeding lines.

Dr. Singh noted that he would be presenting selected data demonstrating how the new genetic materials are catering effectively to diverse ecologies and environmental conditions.

Referring to the recent All India Coordinated Rice Improvement Programme (AICRIP) inaugural session, he drew attention to a new national target introduced by the Director General: a 5% reduction in rice cultivation area and a 10% increase in productivity, termed the "Minus 5, Plus 10" goal. The average rice productivity, currently at 2.7 tons per hectare, is expected to rise to 3.5 tons per hectare. He stressed that this is a serious commitment from the Government of India, calling for enhanced productivity through innovation.

Dr. Singh expressed that the findings from ongoing OFTs will be instrumental in generating evidence to assess which new products are proving effective and how they can be scaled. He reiterated that all stakeholders are working with a shared objective to address climate-related challenges, improve farmers' livelihoods, and support agricultural sustainability.

He also mentioned that, through collaboration with network partners, 33 new genetic lines were released in the previous year, many of which are performing well in decision trials. Moving forward, efforts are being intensified in partnership with the seed systems team to generate robust data and insights from the ongoing OFTs.

Dr. Pradip Dey (Director, ICAR ATARI, Kolkata, Zone-V)

Dr. Pradip Dey opened the session by drawing parallels with insights from the Cereal Systems Initiative for South Asia (CSISA) program, highlighting findings from comprehensive landscape diagnostic surveys conducted in Odisha and West Bengal. He emphasized that despite the availability of improved rice varieties, many farmers continue to grow older varieties, some over 25 to 30 years old, due to various constraints. While these traditional varieties may still perform reasonably well, Dr. Dey pointed out that newer options such as CR Dhan 810 and CR Dhan 811 offer significant advantages in yield and stress tolerance, and could serve as effective replacements for older varieties like Gayatri and Sarala.

He discussed that modern seeds are no longer just inputs for crop production but represent an integrated package, offering benefits such as disease resistance, stress tolerance, and nutritional enrichment, including biofortification. According to Dr. Dey, the cost of not adopting such improved varieties is disproportionately borne by marginalized and poor farming communities, as they miss out on gains in both productivity and nutrition. He advocated for rapid varietal replacement to ensure the timely transfer of these benefits.

Dr. Dey also urged the audience to view the seed sector through a value chain lens, rather than limiting it to the supply chain framework. While institutions like IRRI and CRRI have already established the upstream elements of the chain, such as inbound logistics and processing, attention now needs to focus on downstream components, including institutional development, data interoperability, marketing, and feedback mechanisms that can strengthen value delivery and support decision-making at all levels.

He proposed a shift toward landscape-based varietal targeting, advocating for the clustering of districts based on agroecological conditions such as soil fertility, water availability, and biotic and abiotic stress profiles. By categorizing varietal recommendations according to landscape profiles rather than administrative boundaries, Dr. Dey suggested that more tailored and effective seed packages could be developed.

In conclusion, Dr. Dey emphasized the urgency of accelerating varietal turnover and modernizing seed system strategies. He noted that while progress has been made, a more integrated and rapid approach is essential to fully realize the potential of agricultural innovations and ensure equitable benefits for farmers across diverse ecosystems.

Dr. Sateesh Pareek (R&D Lead, Advanta Seeds, India)

Dr. Sateesh Pareek reflected on the disconnect between the availability of promising new rice varieties and the persistently low varietal replacement rates (VRR) in India. Referencing Dr. Vikas's mention of 33 new genetic lines ready for scale-up, he raised a critical question: why is the adoption of these improved genetics still lagging in a country with a long-standing agricultural legacy?

He commended Dr. Pradip Dey's point on the need to strengthen institutions across the rice value chain, emphasizing that, unlike crops such as millets which require a deliberate value chain build-up, rice already enjoys a well-established system of production, distribution, and end-use diversification. Given this inherent advantage, the focus must now shift toward reinforcing the institutional frameworks that can truly deliver impact at scale.

Dr. Pareek also pointed out the underrepresentation of private sector contributions in rice varietal development and dissemination. He questioned the gaps in India's intellectual property (IP) infrastructure that discourage private companies from openly associating with or investing in public-bred varieties. He raised concerns about the ongoing sale of rice varieties by some private entities without clear attribution or licensing, pointing to the need for stronger regulatory frameworks and incentives to encourage transparent partnerships.

He argued that serious private sector institutions engaged in advanced R&D, including molecular genetics and multi-year breeding programs, should be recognized and supported differently than entities primarily involved in downstream sales. He called for a more enabling ecosystem that promotes innovation, safeguards IP, and rewards genuine research efforts.

Dr. Pareek concluded by emphasizing the importance of not just including private institutions as partners in extension and dissemination, but also in co-developing technologies and varietal pipelines. He welcomed the opportunity to share his perspective and expressed hope for future forums that would deepen dialogue and build effective public-private partnerships in the seed sector.

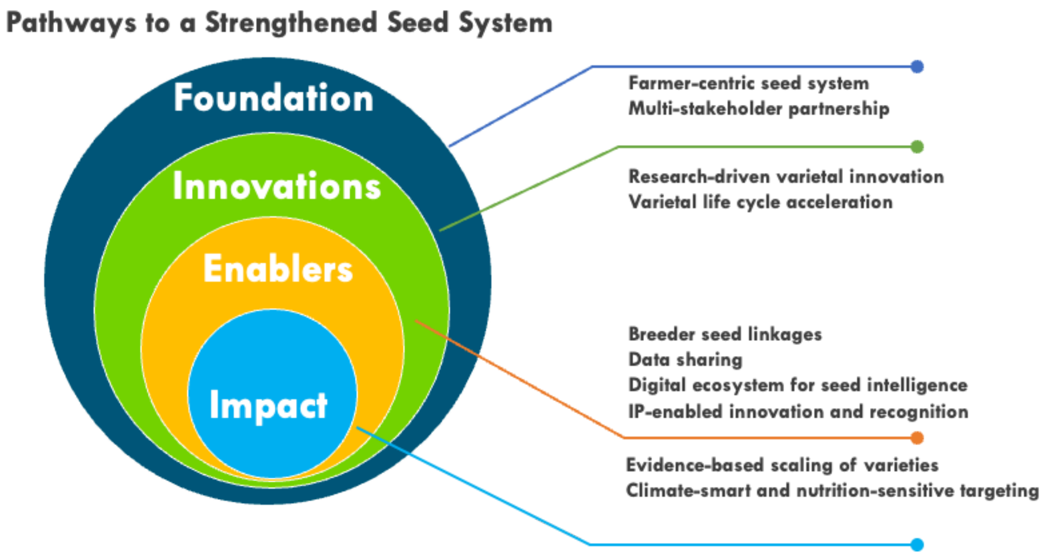

Figure 1: Strategic insights from the opening session

Key Product Diary, Newly Released Varieties Brochure and EGS-Linkage Form

During the event, participants received the latest edition of the Key Product Diary: Rice (India). This comprehensive resource features 30 top-performing rice varieties developed and nominated by Indian breeding institutions. Selected from a pool of 69 varieties, these entries demonstrated superior performance in OFTs conducted between 2022 and 2024 across 1,215 sites in seven states and eight market segments. The varieties consistently outperformed other test entries, benchmark varieties, and farmers' checks.

Each product page within the diary provides detailed information on parentage and the breeding institution responsible for the variety. Importantly, all featured varieties are considered scale-ready for identified agro-climatic regions, providing a practical resource for seed multipliers, FPOs, and other value chain actors.

Additionally, IRRI distributed a brochure listing the rice varieties officially released in 2024 through India's Central and State Variety Release Committees (CVRC and SVRC). These 29 varieties, developed through the IRRI-NARES (National Agricultural Research and Education System) collaboration, incorporate parent lines from IRRI and are tailored to address a range of agro-ecological and production challenges.

Both the Key Product Diary and the Newly Released Varieties Brochure are valuable knowledge products designed to inform breeders, scientists, and stakeholders about newly released varieties, highlighting their potential for enhancing productivity, resilience, and nutritional security. Developed through systematic multi-location testing and stakeholder engagement, these resources aim to support productivity gains, climate resilience, and nutritional outcomes. Following their rigorous on-farm testing through IRRI's systematic evaluation framework, the highest-performing varieties will be prioritized for large-scale dissemination and early-generation seed multiplication, ensuring rapid adoption by farmers.

To strengthen the early-generation seed (EGS) pipeline and streamline varietal scaling efforts, the Seed Accelerator Meet 2025 continued its emphasis on structured seed system linkages. Recognizing that the infusion of EGS into the seed value chain remains critical yet uneven, the meeting provided a renewed platform for institutional coordination and demand articulation.

As part of this initiative, participating seed agencies and stakeholders were provided with an EGS Linkage Support Form. This tool allows them to formally express interest in specific quantities of breeder seeds for high-performing, scalable varieties featured in the Key Product Diary. By facilitating direct linkages between breeding institutions and seed multipliers, the form supports timely access to quality EGS and contributes to a more responsive and demand-driven seed system.

The Key Product Diary, Newly Released Varieties Brochure and EGS Linkage Support Form are included in Appendices 8, 9, and 10, respectively, of this report for reference.

Technical Sessions and Thematic Presentations

Following the inaugural session and opening remarks, the workshop transitioned into a series of thematic and technical presentations. These sessions were led by experts from IRRI, offering in-depth insights into breeding innovations, seed system strategies, varietal deployment, and farmer-centric approaches. Key highlights from each presentation are outlined below:

Market-driven OneRice breeding strategy at IRRI – Dr Vikas Kumar Singh

Dr. Vikas Kumar Singh presented IRRI's strategic advancements and its evolving framework aimed at ensuring a sustainable future for rice production. He explained how IRRI's rice breeding strategies have progressively advanced since the 1960s, with a strong emphasis on developing high-yielding, resilient, and high-quality rice varieties suited to a wide range of climatic conditions.

He elaborated on the OneRice Breeding Strategy, introduced in 2020, which marks a shift towards a more integrated and market-responsive breeding approach. This strategy focuses on aligning varietal development with the specific needs of rice farmers, incorporating cutting-edge tools such as marker-assisted selection and genomic selection to enhance breeding precision and efficiency.

Dr. Singh asserted the importance of market segmentation in guiding breeding priorities, ensuring that targeted traits and ecological needs are addressed effectively. He noted that collaboration with NARES plays a critical role in expanding the reach and impact of IRRI's work, especially in testing and deploying germplasm resilient to both biotic and abiotic stresses. He also shared the significance of the SpeedBreed facility established in IRRI South Asia Regional Centre (ISARC), Varanasi, which has dramatically shortened the rice breeding cycle, from the traditional six to seven years to just one and a half years, thereby accelerating varietal development and delivery.

Key Insights shared by Dr Vikas include:

Evolution of Breeding Techniques: Dr. Vikas traced IRRI's breeding journey from the early development of semidwarf, high-yielding varieties to the current integration of advanced genomic tools. This progression reflects IRRI's adaptive response to evolving agricultural challenges and technological opportunities.

Market-Driven Breeding through Segmentation: He emphasized the importance of detailed market segmentation based on geography, agroecology, cultivation practices, and grain types to guide targeted breeding efforts. This strategy ensures that new rice varieties align not only with productivity goals but also with farmer needs and consumer preferences, enhancing both adoption and market access.

Prioritization of Stress Resilience: A critical focus of IRRI's breeding programs has been the incorporation of traits that address biotic (pests and diseases) and abiotic (drought, flood, heat) stresses. Dr. Vikas highlighted drought-tolerant varieties such as Swarna Shakti Dhan as examples of climate-resilient solutions developed in response to increasing environmental volatility.

Strengthening Collaborations with NARES: He drew attention to the value of IRRI's partnerships with NARES in enhancing research relevance and scaling impact. Joint testing protocols and shared resources facilitate seamless translation of research into practice, improving varietal reach and farmer adoption.

Sustainable Intensification through the OneRice Strategy: He reiterated that the OneRice Breeding Strategy integrates goals of productivity enhancement with environmental stewardship. This dual emphasis supports a more balanced and sustainable approach to rice cultivation, promoting ecosystem health while meeting production demands.

Data-Driven Impact of Modern Breeding: Finally, Dr. Vikas presented evidence that one-third of varieties released since 2020 contain IRRI parentage, showcasing the effectiveness and global relevance of IRRI's updated breeding framework. This trend proves the institution's leadership in modern rice improvement and its role as a model for collaborative agricultural innovation.

Germplasm sharing networks of IRRI for impact acceleration – Dr. Linga Reddy Gutha

Dr. Linga Reddy presented an overview of IRRI's germplasm sharing networks and technology transfer initiatives, emphasizing strategic approaches to intellectual property (IP) management, public-private collaboration, and the promotion of agricultural innovations. He outlined IRRI's licensing frameworks, collaborative partnerships, and project outcomes that collectively aim to deliver scalable impact across diverse rice agri-ecosystems.

Dr. Reddy also explained that IRRI's Technology Transfer Office plays a central role in facilitating innovation dissemination while safeguarding IP, operating under CGIAR's commitment to ensure global public good. A key component of this system is the International Treaty on Plant Genetic Resources for Food and Agriculture (ITPGRFA), which enables equitable access to genetic materials through the Standard Material Transfer Agreement (SMTA), a mechanism that ensures transparency and fairness in germplasm distribution. He mentioned two flagship initiatives: Network for Accelerated Rice Varieties for Impact (NARVI) and the Hybrid Rice Development Consortium (HRDC) as successful models of collaboration that have significantly contributed to varietal innovation, access to new germplasm, and enhanced support to the hybrid rice sector. These platforms, he noted, are instrumental in fostering inclusive partnerships and addressing bottlenecks in varietal development and commercialization.

He further stressed that IRRI's IP strategy is built on principles of fair benefit-sharing, transparent licensing mechanisms, and structured stakeholder engagement, all of which are essential for sustaining innovation ecosystems and maximizing uptake among both public and private sector actors.

The following critical observations were shared by Dr Linga.

Innovative Infrastructure: IRRI's strategic management of IP and collaboration enhances innovation pathways. By fostering partnerships, particularly with private sector entities, IRRI creates a supportive environment where stakeholders can contribute and benefit from advancements in rice technology. This model ensures a continual influx of innovative solutions adapted to evolving agricultural challenges.

Global Compliance Framework: The adherence to the ITPGRFA and the establishment of SMTA reflect a comprehensive approach to genetic resource management. This framework not only supports equitable access but also reinforces the importance of accountability and transparency in germplasm exchange. By following these guidelines, IRRI mitigates legal and ethical risks associated with genetic resource utilization.

Commercial Licensing Strategy: The commercial licensing model described for hybrid rice showcases a structured and transparent process that benefits both IRRI and its partners. Through a tiered approach, commercial entities can obtain valuable genetic resources while supporting the continued availability of innovative breeding materials. This symbiotic relationship drives growth within the sector.

Collaboration as Key to Growth: Partnerships fostered through NARVI and HRDC harness the collective expertise of multiple stakeholders, driving the advancement of hybrid rice technology. These collaborations suggest the need for continuous engagement among researchers, industry leaders, and farmers to diagnose challenges and devise practical solutions rapidly.

Challenges in the Hybrid Rice Sector: The challenges faced by the hybrid rice sector, such as diminishing returns and high R&D costs, signifies the need for innovative models like the Holden Foundation Seed system. Such models present collaborative pathways to share costs and risks associated with hybrid development, promoting sustained investment in research.

Equitable Benefit-Sharing Practices: IRRI's commitment to compliance and equity through benefit-sharing practices is a responsible approach to resource utilization. By ensuring that benefits resulting from research and development are equitably shared, IRRI builds trust within its partnerships and paves the way for sustainable agricultural practices that benefit all stakeholders involved.

Integrating seed system research and inclusive delivery efforts for enhanced breeding impact – Dr. Swati Nayak

Dr. Swati Nayak presented on the complexities and challenges associated with enhancing the impact of breeding through integrated seed systems research. She emphasized the importance of creating an enabling environment to accelerate the adoption of improved rice varieties across India. Her presentation articulated the need for structured, evidence-based strategies to address existing barriers, strengthen linkages across the seed value chain, and ensure that agricultural progress is inclusive, scalable, and beneficial to all stakeholders, from breeders and seed producers to farmers and consumers.

Dr. Nayak explained the critical role of integrated seed systems in maximizing the impact of breeding programs by bridging the gap between varietal innovation and adoption. She noted that these systems must be inclusive and strategically aligned to serve underrepresented groups, particularly women, youth, tribal communities, and smallholder farmers. Addressing existing adoption challenges, Dr. Nayak stressed the importance of evidence-based approaches such as on-farm trials, digital evaluation tools, and participatory varietal selection to ensure that new varieties align with both agronomic performance standards and farmers' preferences.

She suggested a shift towards demand-driven varietal development, prioritizing high-yielding, climate-resilient, and nutritionally enriched rice varieties tailored to specific market segments. In this context, IRRI's integrated research efforts in India were presented as a model, designed to enhance seed replacement rate (SRR) and improve the reach of improved varieties. Dr. Nayak identified key barriers to adoption, including inadequate varietal positioning, limited farmer awareness, and restricted access to quality seeds. However, she noted that certain regions, such as Odisha, have demonstrated encouraging varietal turnover trends, showcasing effective integration of IRRI's strategies and providing scalable models for other regions.

In addition, she highlighted regional initiatives like the Seeds Without Borders, which facilitates cross-border sharing of successful rice varieties to improve genetic access, biodiversity, and food security. This initiative, she said, exemplifies the value of regional cooperation in enhancing agricultural productivity and resilience.

Overall, Dr. Nayak concluded that building inclusive, participatory, and data-driven seed systems is fundamental to transforming rural livelihoods and achieving sustainable agricultural development.

Key findings from OFTs and seed scaling strategy – Dr Mosharaf Hossain

Dr. Mosharaf Hossain delivered a comprehensive presentation outlining the OFTs conducted by the IRRI Seed Systems team in the last three years (2022-24), with a particular focus on varietal performance evaluations and community-led seed amplification initiatives. He offered key insights into the current dynamics of rice productivity in India, sharing national trends in varietal adoption and introducing the concept of Weighted Average Varietal Age (WAVA) as a critical metric to assess varietal turnover and the adoption rate of newer, improved rice varieties.

While there has been a gradual improvement in rice productivity across the country, Dr. Hossain noted that challenges remain, particularly the slow adoption of improved varieties. He pointed out that India's rice productivity has seen a modest compound annual growth rate (CAGR) of 2.2%, indicating incremental gains in agronomic practices and varietal improvements. Despite the release of over 1,518 rice varieties and hybrids, developed and approved by both central and state variety release committees for more than 18 ecosystems, the rate of adoption continues to lag. Key barriers cited include farmer concerns related to yield stability, disease resistance, and nutritional quality.

To address these adoption gaps, Dr. Hossain advocated for participatory approaches such as varietal cafeterias, on-farm trials, and cluster demonstrations, which not only validate varietal performance in real-world conditions but also enhance farmer engagement and informed decision-making. He emphasized the importance of segmentation strategies, grounded in agro-ecological and socio-economic data, to support localized varietal deployment and agronomic recommendations tailored to specific environments and farmer preferences.

Dr. Hossain disussed the use of advanced statistical and analytical tools, including R programming and data visualization platforms, to assess varietal performance more accurately. This data-driven approach is central to IRRI's continuous learning framework, ensuring adaptive varietal management strategies responsive to evolving challenges in rice ecosystems. He flagged that extending OFT assessments beyond 2024 would be key to refining breeding pipelines and bridging information gaps through stakeholder-driven research.

He provided a detailed overview of the wet season OFTs in 2024, during which 180 trials were conducted across 11 districts in four Indian states -Odisha, Uttar Pradesh, Chhattisgarh, and Jharkhand, in collaboration with 14 NARES partners, 5 KVKs, and 6 NGOs. A total of 31 test varieties were evaluated against 8 benchmark and popular farmer varieties, across five key market segments, and the best-performing varieties were subsequently identified as follows:

- TEMS-I: CR Dhan 320, NLR 3238
- TLaSF-R: MTU 1321
- TMeLS-R/I: Ratnagiri 8, RNR 29325
- DELS-R/I: Arize 6129 Gold, AZ 6585 ST, AZ 8455 DT, PR 126
- DMeLS-R/I: AZ 8455 DT, CR Dhan 602

A detailed analysis of the best-performing varieties from 2022 to 2024, including associated yield gains, is provided in Appendix 7*.*

Dr. Hossain also discussed IRRI's broader stakeholder engagement strategy, emphasizing its role in strengthening the resilience and sustainability of rice-based systems. In particular, he stressed the importance of community participation in seed systems, elaborating gender-sensitive capacity-building efforts aimed at empowering FPOs, FPCs, NGOs, and individual seed growers across Uttar Pradesh, Assam, and Odisha. These efforts have enabled the local production of foundation, certified, and truthfully labelled seeds (TLS) for a range of improved varieties such as LR 3354, PR 126, Telangana Sona, Ratnagiri 8, DRR 48, DRR 53, DRR 56, BRRI Dhan 84, BRRI Dhan 100, Swarna Sub 1, Protizen, and Zinco Rice MS.

Additionally, IRRI is working in close coordination with state seed corporations in Bihar, Chhattisgarh, Odisha, and Uttar Pradesh to support seed scaling initiatives, ensuring that promising rice varieties are multiplied and made accessible to farmers at scale.

Special Talk - Dr. Mahalingam Govindaraj

As part of the Seed Accelerator Meet, a special address was delivered by Dr. Mahalingam Govindaraj, Senior Scientist with the HarvestPlus program, on the theme of “Scaling and Commercialization of Specialty and Healthier Rice Segments”, including biofortified, low glycemic index (GI), and high-protein rice. His insights were particularly relevant to the scaling partners present, given the emerging opportunities in the nutrient-rich rice sector and the critical aspects of positioning and marketing such varieties.

Dr. Govindaraj began by thanking the organizers for facilitating an engagement platform that brought together a diverse group of stakeholders. He emphasized that his work focused on the development and promotion of biofortified rice varieties, such as zinc-enriched rice, in collaboration with institutions including the International Rice Research Institute (IRRI) and the International Centre for Tropical Agriculture (CIAT).

He explained that the nutrient density of crops, particularly micronutrients, had been declining due to various biotic and abiotic stresses. HarvestPlus adopted an incremental approach to address this, identifying baseline nutrient levels for different crops. For instance, he noted that the global baseline for zinc content in rice is 16 ppm, while some untargeted breeding efforts have already resulted in rice mega-varieties containing over 18 ppm zinc. However, such progress has not been uniform across all crops. Dr. Govindaraj cautioned that focusing exclusively on yield can sometimes lead to nutrient loss, making it essential to simultaneously select for both yield and nutritional traits, particularly protein and zinc, which he noted are highly correlated.

Citing an example from wheat, he mentioned Akbar 2019, a zinc-enriched variety developed in partnership with the Wheat Research Institute in Pakistan. This variety now covers nearly 40% of the wheat-growing area in Punjab, Pakistan. Although adopting a single variety across large areas carries risks, Akbar 2019 has proven popular due to its high yield, heat tolerance, and enhanced zinc content. Dr. Govindaraj urged stakeholders to prioritize the development and commercialization of crop varieties with such novel, nutrition-focused traits, emphasizing that micronutrient enrichment is not just an agronomic trait but a “social trait” critical for achieving nutritional security.

He introduced HarvestPlus Solutions, a for-profit arm of the program that facilitates technology and germplasm exchange between CGIAR institutions, NARES, and the private sector. With operations in five countries, the entity plays a vital role in strengthening agricultural technology transfer and fostering collaborative development of nutrient-rich crops.

Dr. Govindaraj stressed the importance of leveraging staple foods, widely consumed across rural and vulnerable populations, as vehicles for delivering essential micronutrients. He reiterated the urgency of mobilizing resources to reach the HarvestPlus goal of improving the nutrition of one billion people by 2030. He also shared that HarvestPlus focuses on three key micronutrients - iron, zinc, and vitamin A, endorsed by the World Health Organization and considered essential due to their role in addressing anaemia, stunting, and blindness, respectively. These deficiencies are especially prevalent in low- and middle-income countries, including India.

Dr. Govindaraj further elaborated on the Biofortification Priority Index, an open-access tool available online that allows users to select a crop, geography, and nutrient to identify priority areas for investment. This tool considers factors such as native genetic variation, per capita consumption, and the prevalence of nutrient deficiencies in specific regions.

Addressing the growing interest in low glycemic index (GI) crops, he cautioned that focusing exclusively on one trait, such as low GI, should not come at the expense of yield or other nutritional benefits. He reinforced the need to strike a balance between traits such as resistant starch, micronutrient density, and yield, especially in the context of an impending yield plateau in major staples like rice. The pathway forward, he concluded, involves continuous innovation and evidence-based marketing of nutrient-rich varieties, ensuring that value-added traits are substantiated and effectively communicated to consumers and stakeholders alike.

Table 1: Roles and potentials across stakeholder groups

	ICAR ATARI (via KVKs)	IRRI	Private Seed Companies	Community Seed Institutions
Role in Breeding	Alignment with national goals, variety trials under farmer conditions	Breeding, varietal testing, OFTs	Hybrid R&D, advanced molecular breeding, downstream trials	Limited (participate in trials or validation)
Role in Dissemination	Extension support, validation trials, demonstrations	Facilitating seed demand forecasting, early-generation seed supply	Commercial scaling through dealer networks, B2B channels	Local seed production, varietal demonstrations, feedback to breeders
Challenges	Fragmented feedback channels, variable KVK capacities, limited capacity for post-trial scaling	Breeder seed to production scale bottlenecks	IP protection gaps, limited incentives for public variety promotion	Technical capacity gaps, quality assurance, financial sustainability
Opportunities	Strong national reach, integrated with policy ecosystems, field-level evidence generation	Public varietal release pathway	Co-development of high-value traits, hybrid-inbred synergies	Trusted by farmers, agile in varietal promotion, key for niche ecology targeting

Group Discussion: Selection, Deployment, and Scaling of Rice Varieties

As a core component of the Seed Accelerator Meet 2025, a structured group discussion was held on the selection, deployment, and scaling of rice varieties, with a focus on aligning varietal promotion strategies with stakeholder needs and regional priorities. The session aimed to identify actionable pathways to enhance varietal turnover, strengthen last-mile seed delivery, and accelerate adoption of improved varieties across diverse geographies.

The discussion was organized in three stakeholder-specific phases, enabling tailored dialogue and targeted insights from ICAR ATARI and SSCs, private seed Companies and FPOs/FPCs. Key takeaways from each phase of the discussion are summarised below.

Perspectives from ICAR ATARI and state seed corporations

The discussion among state government-level stakeholders brought to light region-specific constraints and systemic challenges in the selection, deployment, and scaling of rice varieties. Officials from SSCs and ICAR institutions shared practical experiences, policy-level bottlenecks, and operational challenges faced in their respective states.

Regional Varietal Needs and Production Constraints in Assam and North-East India

Dr. H.C. Bhattacharya of ICAR ATARI, Guwahati, flagged the need for short-duration rice varieties suitable for jhum cultivation systems in the North-East Indian ecosystem, particularly due to low productivity and shifting cultivation practices prevalent in the region. Despite the presence of a structured seed certification mechanism through the Assam Seed and Organic Certification Agency (ASOCA), a lack of synergy among key institutions, namely ASOCA, the Assam State Seed Corporation, KVKs, and Agricultural Universities, was identified as a major hurdle. Dr. Bhattacharya noted that due to high Minimum Support Prices (MSPs), many farmers opt out of seed production, as the assured grain price acts as a disincentive to participate in seed multiplication programs.

He further revealed the gap in varietal dissemination, observing that promising rice varieties, even after successful On-Farm Trials, often do not reach the farmers due to weak extension linkages. This was supported by Dr. Sudesh Sharma from Signet Seeds, who cited the examples of Pant Dhan 18 and Pant Dhan 19, high-yielding, drought-resilient varieties, that failed to scale despite proven performance in different agro-climatic zones.

Dr. Bhattacharya stressed the need to shorten the seed chain and enhance frontline demonstrations following OFTs. KVKs were recognized as key players in varietal evaluation but were limited in outreach capacity. It was suggested that institutions like IRRI could partner with KVKs to introduce and generate evidence on new varieties. ATARI Guwahati is currently undertaking an initiative to compile data on the best rice technologies demonstrated over the last five years across its KVKs, to strengthen varietal promotion strategies.

Seed Value Chain and Infrastructure Gaps in Assam

Mr. Gunjanan Wallung Shyam of the Assam State Seed Corporation acknowledged that the continued procurement of older rice varieties, such as Ranjit Sub-1 from West Bengal, is primarily due to the non-availability of newer alternatives and weak seed supply chains. He raised concerns regarding the proliferation of spurious seeds and pointed out that government-supported storage infrastructure in Assam caters primarily to grain, not seed.

He added that the absence of community dryers, rodent-proof storage facilities, and post-harvest infrastructure severely limits seed quality maintenance. Most seed growers in Assam are marginal farmers with less than 0.5 acres of land, making access to drying and storage technologies financially unviable. The humid climate of the region exacerbates post-harvest seed quality issues. In this context, Dr. Bhattacharya added that varietal dormancy and duration are intricately linked to agro-climatic conditions, particularly in humid zones.

Structured Procurement and Market Incentives in Uttar Pradesh

Mr. Anuj Kumar Singh from the Uttar Pradesh Beej Vikas Nigam (UBVN) noted that seed production in UP is aligned with projected demands. UBVN produces around 20% of the seeds, while state seed farms account for the remaining 80%. Breeder seed procurement is often carried out through collaborations with IRRI. However, a noticeable shift in farmer preference towards hybrid rice has resulted in declining state-led rice seed production.

To encourage quality seed production, UBVN offers a price incentive of 20–25% above the market rate, or 30% above MSP, to seed growers. It also provides a 50% subsidy to farmers at the time of seed purchase, making it an economically viable proposition for both producers and users.

Policy and Market Disincentives in Odisha

Mr. Aditya Kumar Panda of the Odisha State Seed Corporation (OSSC) pointed out the challenges arising from policy-driven varietal introductions. He noted that multiple varieties are being promoted in Odisha. Many of these varieties fail to proceed beyond the foundation seed stage due to concerns over maintaining quality during large-scale multiplication.

The recent shift to Direct Benefit Transfer (DBT) for seed subsidies has led to a 30–50% decline in OSSC sales. Farmers are now required to pay the full purchase price upfront, with subsidies credited to their accounts after six months. This change has led many farmers to prefer private agri-input dealers, who offer more flexible purchase mechanisms. OSSC continues to rely on its network of approximately 10,000 registered individual seed growers for seed production.

Strengths and Technical Challenges in Maharashtra

Dr. Prafull Lahane of Mahabeej (Maharashtra State Seeds Corporation) reported a rising demand for early maturing rice varieties (110-day duration) with short, slender grains. However, the availability of such seeds remains limited. Given that IRRI is unable to directly supply breeder seeds to interested institutions, Mahabeej is open to facilitating such transfers through Memoranda of Understanding (MoUs). Based on OFT results, Mahabeej also expressed interest in procuring breeder seeds or germplasm for certified seed production.

Mahabeej possesses 23 seed processing units and advanced facilities, including seed testing laboratories equipped with DNA fingerprinting capabilities. The agency has requested capacity-building support in the areas of DNA fingerprinting and seed health testing. Nonetheless, major challenges persist with respect to insect-pest infestations during storage and inadequate fumigation practices. Dr. Lahane suggested that these challenges may be common across other SSCs and proposed a joint discussion to explore possible solutions.

Regional Perspectives in Rice Seed System

Assam & North-East India	Uttar Pradesh	Maharashtra	Odisha
Need for short-duration, jhumsuited varieties Low synergies among key institutions Reliance on older varieties Lack of community drying, storage facilities (humid climate)	Shift in preference towards hybrid rice Huge incentives and subsidies (Price incentive of 20-25% above the market rate to seed growers; 50% subsidy to farmers at the time of seed purchase)	Need for early maturing slender grain varieties Storage pest and fumigation challenges Capacity building support in seed health testing and DNA fingerprinting	Over-promotion of multiple varieties Shift to direct benefits transfer (DBT) for seed subsidies led to a 30-50% drop in OSSC seed sales Failure to progress past foundation seed

Figure 2: Perspectives from ICAR ATARI and state seed corporations

Perspectives from private sector stakeholders

The private seed sector plays a pivotal role in the development, production, and dissemination of rice varieties in India. However, several challenges hinder their operations. It is crucial to address the challenges to meet the evolving needs of Indian agriculture.

Emerging Pest Threats and the Need for Resistant Varieties

Private seed companies have expressed significant concern over the increasing prevalence of pests such as Brown Planthopper (BPH) and gall midge, which pose substantial threats to rice yields. Dr. Rajesh Mishra of Harlal Seeds emphasized the urgent need for rice varieties that are resistant to these pests.

Varietal Suitability for Natural Farming Practices

The shift towards natural farming in Andhra Pradesh has led to a demand for rice varieties compatible with organic cultivation methods. The Andhra Pradesh Community Managed Natural Farming (APCNF) program has been instrumental in promoting natural farming, with over 10 lakh farmers adopting chemical-free practices across the state. Dr. Mishra inquired about the availability of such varieties.

Challenges in Breeder Seed Accessibility

Private seed companies have reported difficulties in accessing breeder seeds from the National Seeds Corporation (NSC). Dr. Mishra pointed out that indents placed by private entities are often not fulfilled timely manner, hindering seed production schedules. Dr. Vikram Yadav of Indo American Hybrid Seeds echoed this concern, emphasizing that while private companies can scale seed production, the unavailability of breeder seeds remains a bottleneck. Addressing this issue, Mr. Anuj Kumar Singh from Uttar Pradesh Beej Vikas Nigam suggested that indents for breeder seeds should be placed well in advance to ensure timely supply, as the government prioritizes such requests to boost certified seed production.

Enhancing Varietal Data Sharing and Positioning

The private sector has expressed the need for more comprehensive data on new rice varieties to aid in effective positioning and marketing. Dr. Mishra recommended that IRRI share trial results, including yield gains over popular and benchmark varieties, to assist private companies in varietal selection. Additionally, Dr. Sudesh Sharma of Signet Seeds inquired about the possibility of testing company-released varieties through IRRI's On-Farm Trials. Dr. Swati Nayak responded that currently, only products nominated by NARES partners are considered for such trials, suggesting a potential area for future collaboration.

Leveraging Surplus Seed Production

To address seed shortages, Dr. Mishra proposed that seed growers associated with SSCs corporations could produce surplus seeds beyond the indented quantities and supply them to private companies. This approach could enhance seed availability and foster public-private partnerships in seed distribution.

The insights from private sector stakeholders signify the critical need for collaborative efforts between public institutions and private entities to address challenges in pest resistance, seed accessibility, and data sharing. Such partnerships are essential for the effective selection, deployment, and scaling of rice varieties that meet the evolving needs of farmers and the agricultural sector at large.

Strengthening the Rice Seed Ecosystem

Biotic Stress Challenges

Emerging pest threats
Need for resistant variesties

Agro-ecological Alignment

Varietal performance date for positioning insights
integration of private varieties in OFTs

Seed Supply Chain Gaps

Challenges in timely delivery of breeder seeds

Data & Evidence Needs

Varietal performance data for positioning insights
Integration of private varieties in OFTs

Figure 3: Perspectives from private seed companies

Perspectives from community-level seed institutions (FPOs/FPCs)

FPOs and FPCs are pivotal in enhancing the accessibility of quality rice seeds at the grassroots. However, stakeholders raised concerns on several on-ground challenges that restrict the efficiency and sustainability of these organizations in the rice seed sector.

Need for Standard Operating Procedures (SOPs) in Seed Production

Mr. Satyabrat Biswal from the Agro Ecology Society called attention to the necessity for comprehensive SOPs at various stages of seed production to assist FPOs in maintaining seed quality. According to him, FPOs often lack the technical guidance necessary for producing high-quality seeds. SOPs could serve as structured manuals that inform seed growers about critical practices - foundation seed procurement, field inspections, processing, storage, and distribution.

Challenges in Foundation Seed Availability and Production

Access to foundation seeds remains a significant hurdle for FPOs. Mr. Biswal pointed out that the availability of the right varieties or foundation seeds is a persistent bottleneck. Overestimation of demand can lead to surplus production, resulting in financial losses and wastage. Conversely, underestimation or unforeseen production failures, owing to factors like climate variability, negligence, pest attacks, or disease outbreaks, can cause a seed shortfall, leaving dealers and farmers without sufficient supply during the sowing season. So, demand forecasting is a big challenge.

Storage Infrastructure Deficits

Storage and processing infrastructure also emerged as a major area of concern. As Mr. Biswal stated, many FPOs are forced to rely on rented seed processing units, which may not meet scientific storage standards. This adds to their costs and affects quality control. He advocated for a comprehensive SOP that not only outlines production methods, but also covers inventory planning and scientifically sound storage practices, which could prevent avoidable post-harvest losses and preserve seed viability.

In Assam, Mr. Ranjan Kr Barthakur of Poohar Agro Producer Company shared the limitations in post-harvest processing and community-level infrastructure. He shared that during the Boro season, which is important for rice seed production in Assam, frequent and untimely rains make it difficult to maintain seed quality. As a result, seed production during this period is almost unviable. He suggested that in such cases, foundation seeds could be provided to FPCs or FPOs in neighbouring states where climatic conditions are more favourable for seed multiplication. This approach could help optimize the regional strengths and maintain seed availability.

Market-Specific Varietal Promotion

Mr. Shardool Vikram Chaudhari from Jaya Seeds brought attention to the need for market-segment-specific promotion of rice varieties. According to him, it is not enough to develop a high-yielding or pest-resistant variety; the success of a variety also hinges on its acceptability by local millers and grain traders. If a variety does not align with the grain quality preferences of the processing and retail segments, farmers are likely to reject it in subsequent seasons. He advocated for a well-defined process that follows on-farm trials with milling assessment, grain quality analysis, and then strategic varietal positioning. The variety's success, he stressed, ultimately depends on the perceptions and experiences of farmers, which must be considered while designing dissemination strategies.

Financial Constraints and Pricing Challenges

On the financial front, Mr. Raja Shekhar from the Centre for Sustainable Agriculture (CSA) stated that working capital or operational funds remain a chronic constraint for FPOs. Without adequate upfront funds, FPOs struggle to procure inputs, pay for storage, or cover processing costs in a timely manner. In Telangana, he noted that apart from the centrally declared MSP there is an additional bonus for fine varieties, which influences market pricing. FPOs under CSA are facing difficulties in deciding optimal pricing strategies that balance both affordability for farmers and sustainability for the FPOs. Striking this balance is critical to ensure that farmers are willing to purchase from FPOs instead of going to private retailers or local traders.

While FPOs and FPCs hold immense potential to revolutionize rice seed systems in India, addressing the outlined challenges is imperative. Implementing standardized procedures, ensuring the timely availability of foundation seeds, enhancing storage infrastructure, aligning varietal promotion with market demands, and alleviating financial constraints will collectively empower these organizations to better serve the farming community.

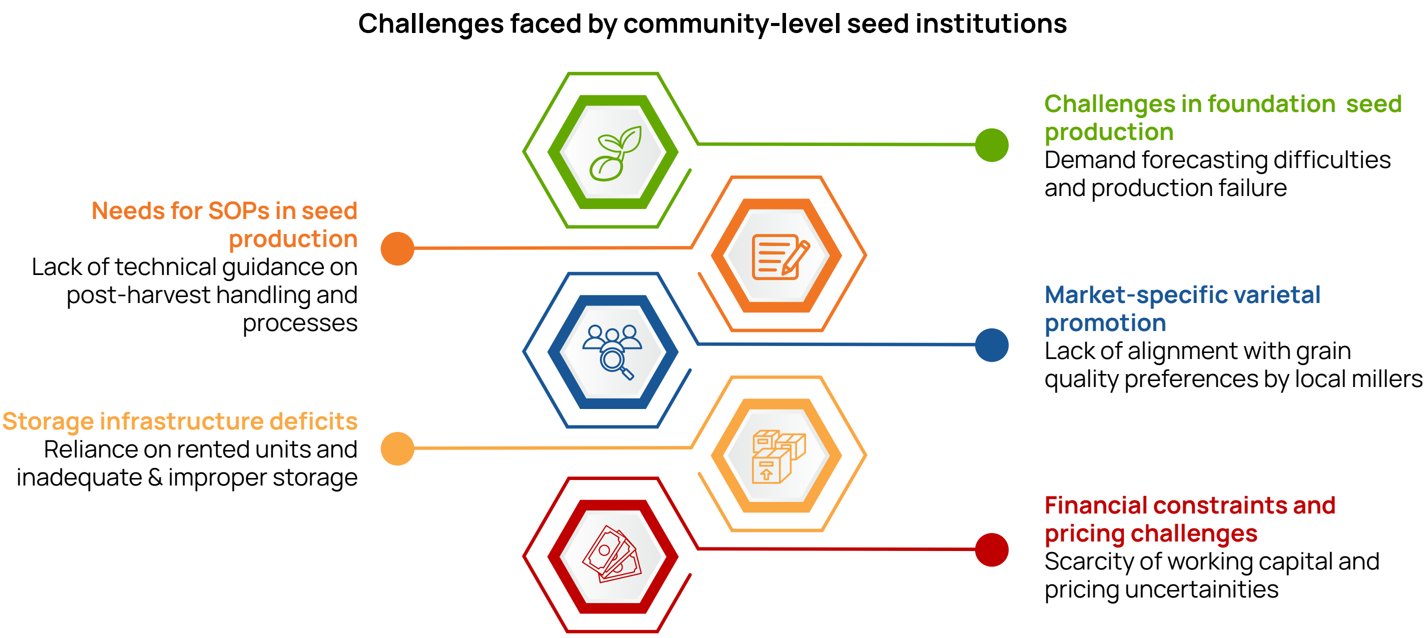

Figure 4: Perspectives from FPOs/FPCs

Conclusion

The Seed Accelerator Meet 2025 served as a dynamic convergence point for actors across the Indian seed ecosystem, ranging from government institutions and private seed enterprises to FPOs and NGOs. This year's deliberations emphasized actionable partnerships, evidence-based scaling of superior varieties, and agile seed systems that respond to climate, market, and nutritional priorities.

A key takeaway was the urgent need to bridge gaps between innovation and impact by streamlining the flow of information, seeds, and feedback between breeders and farmers. Participants discussed the importance of demand forecasting, tailored varietal promotion strategies, and decentralized seed production models that empower local actors. The conversations reflected a maturing seed ecosystem, one increasingly driven by inclusivity, resilience, and evidence-backed decisions.

As the event concluded, participants committed to developing decentralized varietal pipelines, enhancing last-mile varietal access, and deepening private-public-civil society partnerships. The meet reinforced that the future of India's seed systems hinges not only on innovation, but on intentional, cross-sector collaboration and shared accountability.

Appendices

Appendix 1: List of Participants

Sl. no.	Name	Designation and organization
1	Dr. Pradip Dey	Director, ICAR ATARI, Kolkata (Zone-V)
2	Dr. H.C Bhattacharya	Scientific Consultant, ICAR ATARI Guwahati, Zone VI
3	Dr. Shantanu Kuar Dubey	Director, ICAR ATARI, Kanpur (Zone-IV)
4	Mr. Anuj Kumar Singh	Consultant, Uttar Pradesh Beej Vikas Nigam
5	Mr. Gunjanan Wallung Shyam	Asst. Branch Manager, Assam Seeds Corporation
6	Mr. Aditya Kumar Panda	DGM, Odisha State Seed Corporation
7	Dr. Prafull Lahane	GM, Quality Control, Mahabeej, Akola
8	Mr. Vivek Thakare	GM, Production, Mahabeej, Akola
9	Mr. Shardool Vikram Chaudhari	Director Production, Jaya seeds, Varanasi
10	Mr. Manoj Kumar	Functional Head, R S Seeds, Varanasi
11	Mr. Ranjan Kr Barthakur	Director & Chairman, Poohar Agro Producer Company Ltd, Guwahati
12	Mr. Satyabrat Biswal	CEO, Agro-ecology Society, Odisha
13	Mr. Abhay Sanker Tiwari	Functional Head, Navpragyan FPC, Varanasi
14	Dr. Raja Shekhar G	COO, Centre for Sustainable Agriculture, Hyderabad
15	Dr. Prabhugouda Patil	Senior Breeder, Advanta Seeds, Hyderabad
16	Dr. Satish Pareek	R&D Lead, Advanta Seeds, Hyderabad
17	Mr. Vishal Kumar	CEO, Atamanda Seeds, Hyderabad

S. no.	Name	Designation and organization
18	Mr. Md. Sajid	Director Sales, Atamanda Seeds, Hyderabad
19	Dr. Amaresh Chandel	Breeding and Partnership Lead, Bayer Crop Science, Hyderabad
20	Dr. Rajesh Mishra	Director Research, Harlal Seeds, Hyderabad
21	Dr. Vikram Yadav	Senior Breeder, Rice, Indo-American Hybrid Seeds, Hyderabad
22	Dr. Sharatbabu Sonnad	Crop Lead - Rice, Mahindra Agri Solutions, Hyderabad
23	Dr. P. Ravi Yugandhar	Rice Breeder, Mahindra Agri Solutions, Hyderabad
24	Dr. Devender Kumar Kadian	Head -Breeding, Nuziveedu Seeds, Hyderabad
25	Dr. Kuldeep Tyagi	Director Research, Signet Seeds, Hyderabad
26	Dr. Sudesh Sharma	MD, Signet Seeds, Hyderabad
27	Dr. Gagandeep Singh	Senior Breeder, Rice, Star AgriSeeds, Sangaria
28	Dr. Amit Shukla	DGM Research, Trimurti Plant Science, Hyderabad
29	Dr. Himanshu Tyagi	Director, Vertical AgroSciences, Noida
30	Dr. Sayed Akhter	Director Sales, Eagle Seeds, Hyderabad
31	Dr. Avinash Umate	Assistant Breeder (Paddy), VNR Seeds, Hyderabad
32	Dr. Subhash C. Yadav	Rice Breeder, Eldorado Seeds, Hyderabad
33	Dr. Srijib Panda	Senior Rice Breeder, Siri Seeds, Hyderabad
34	Dr. Sunil Naik	Crop Breeding Lead, Rallis India, Hyderabad

Appendix 2: Program Agenda of the Workshop

S. No	Subject	Session Lead	Schedule
Registration			9.00 – 9.30
1	Overview of programme and welcome remarks	Dr Swati NayakProgramme Convenor	9.30 – 9:40
2	Opening ceremony and special remarks by key guests		9:40 – 10:40
3	Overview of market driven One Rice Breeding strategy at IRRI	Dr Vikas Kumar Singh	10.40 – 11:10
4	IRRI's germplasm sharing network and its modalities	Dr Linga Reddy Gutha	11:10 – 11:30
5	Integrating seed system research and inclusive delivery efforts for enhanced breeding impact	Dr Swati Nayak	11.30 – 12.00
6	On-farm testing and result sharing for new products/varieties (Evidence from last three-year adaptive trials)	Dr Sk Mosharaf Hossain	12.00 – 12.40
7	Scaling and commercialization of specialty and healthier rice segments	Dr Mahalingam Govindaraj	12:40 – 13:00
Lunch Break			13:00 – 14:00
8	Open discussions and potential R&D linkages and partnerships (Sharing of request forms for EGS linkage support)	Facilitated by Dr Neeraj Tyagi	14:00 – 14:30
9	Group Discussion: Selection, deployment, and scaling of varieties (Rice Case)	Moderated by Dr Swati Nayak	14.30 – 16:30
10	Closure	Dr Swati Nayak	16.30 – 16.40
Tea and Disperse			16.40 – 17.00

Appendix 3: Global Market Segments (Rice Cultivation)

S. no.	Market segment	Maturity segment	Features
1	TEMS-I	Early	Transplanted early-duration medium slender soft grain rice for irrigated ecosystem
2	DELS-I	Early	Direct-seeded, early-duration, long slender soft grain rice for irrigated ecosystem
3	DELS-R	Early	Direct-seeded, early-duration, long slender soft grain rice for upland shallow areas
4	DEMS-R	Early	Direct-seeded, early-duration, medium slender soft grain rice for rainfed ecosystem
5	TMeLS-I	Medium	Transplanted medium-duration, long slender soft grain type rice for irrigated areas
6	DMeLS-R	Medium	Direct-seeded, medium-duration, long slender soft grain type rice for rainfed areas
7	DMeLS-I	Medium	Direct-seeded, medium-duration, long slender soft grain type rice for irrigated areas
8	TLaMF-I	Late	Transplanted, medium-late-duration, medium slender, firm and dry grain type rice for irrigated areas
9	TLaSF-R	Late	Transplanted, late-duration, small, firm grain type rice for rainfed areas

Appendix 4: Presentation by Dr. Vikas Kumar Singh

Appendix 5: Presentation by Dr. Linga Reddy Gutha

Appendix 6: Presentation by Dr. Swati Nayak

Appendix 7: Presentation by Dr. Mosharaf Hossain

Appendix 8: Key Product Diary

Appendix 9: Newly Released Varieties Brochure

Appendix 10: Early Generation Seed (EGS) Linkage Support Form for Seed Agencies

IRRI collaborates with the NARES network to implement structured on-farm testing of newly released rice genotypes across Southeast Asia, ensuring rigorous evaluation under local ecological conditions. Varieties proven well-adapted through these OFTs are prioritized for specific ecologies and market segments. To support rapid dissemination of these varieties, IRRI facilitates the flow of Breeder and Foundation seeds through NARES partners to community-based organizations (FPOs, FPCs, SHGs) and private seed companies, thereby enhancing varietal replacement and seed replacement rates and expediting farmer adoption of improved rice varieties.

Requester Details

Name of Requester:	
Designation:	

Organization/Company Details

Name of Organization/Company:	
Address:	
Phone:	
Email:	
Website Link:	

Short Introduction of the Organization

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(Please complete all fields)

Sl. No	1	2	3
Rice Variety Required			
Class of Seed (BS/FS)			
Seed Quantity (kg)			
Required By (e.g., by mid-May)			
Company Has Required Licensing for Seed Marketing (Yes/No)			
Location for Seed Multiplication			
Targeted Area/Ecology for Dissemination /Marketing			